

# **Advanced Materials „Soft Matter“**

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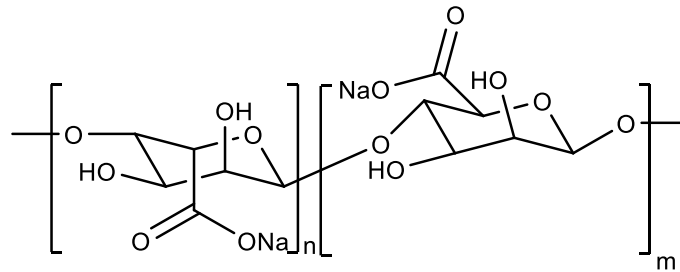
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## Practical Course

- 1 Synthesis of Physically Cross-linked Gel-Building Polymer
- 2 Synthesis of Chemically Cross-linked Gel-Building Polymer
- 3 Synthesis of poly-(N-isopropylacrylamide) microgels via precipitation polymerization
- 4 Synthesis of poly-(N-isopropylacrylamide)-co-methacrylic acid microgels via precipitation polymerization

## 1. *Synthesis of non-Covalently Cross-linked Polymer Gel*<sup>1</sup>



Of all the different biopolymers that can be employed in the formation of the semipermeable membrane in capsules, alginate is one of the most frequently used owing to the fact that its immobilization is carried out under very mild conditions. In molecular terms, alginate consists of a family of unbranched binary copolymers of (1-4)-linked  $\beta$ -D-mannuronic acid (M) and  $\alpha$ -D-guluronic acid (G) of widely varying composition and sequential structure. Alginates are true block copolymers composed of homopolymeric regions of M and G, termed MM and GG blocks, respectively, interspersed with regions of alternating structure (MG blocks). In solution, alginates behave like flexible coils. However, upon interaction with divalent metal ions, such as  $\text{Ca}^{2+}$ , they form an ordered structure. Grant et al. showed that the gelation process involves cooperative binding of calcium ions between aligned GG blocks of two alginate chains. The gelation of calcium alginate is an almost instantaneous and irreversible process, which is governed by the relative rate of diffusion of calcium ions and polymer molecules into the gelling zone. This fact indicates that the gelation process can be accurately expressed by the relationships used for other diffusion-limited reaction systems.

### Materials

Sodium alginate  
Calcium chloride  
Distilled water

Water bath  
Beaker flask (3 x 100 mL)  
Büchner funnel  
Filter paper  
Magnetic stirrer (with magnetic stir bar)  
Magnetic stirrer (with contact thermometer and magnetic stir bar)  
Syringe (2 x 2,5 mL)

<sup>1</sup> Ana Blandino, Manuel Macias, Domingo Cantero, Formation of Calcium Alginate Gel Capsules: Influence of Sodium Alginate and  $\text{CaCl}_2$  Concentration on Gelation Kinetics, Journal of Bioscience and Bioengineering (1999), Vol. 88, No. 6, 686-689.

1) In a water bath, a beaker flask with 50 mL water is heated at about 90°C. Then sodium alginate (200 mg) is dissolved in the hot water.

2) In a second and third beaker flask, calcium chloride is dissolved in water.

3) **Alginate particles:** The sodium alginate solution is dropped (about 10 drops) through a syringe into the calcium chloride (50 mL, 0,5 M). It can be observed, that sodium alginate builds a white precipitate (globular particles) in calcium chloride solution. These are taken out and the water is removed with a syringe.

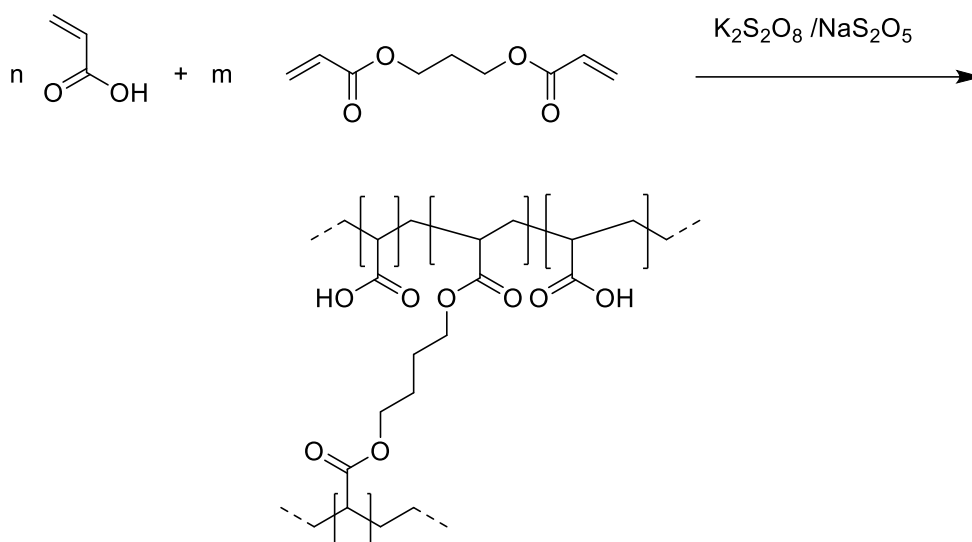
4) **Alginate capsules:** Droplets of calcium chloride solution (50 mL, 1 M) are added dropwise (about 5 drops) through a tube into sodium alginate solution, using a syringe. A dropping height of about 10 cm ensures that spherical droplets are formed. The sodium alginate solution is constantly stirred at 350 rpm using a magnetic stirrer situated at the bottom of the vessel, in order to keep the droplets from sticking together and to minimize external transfer resistance.

Once the  $\text{CaCl}_2$  solution was added dropwise into the alginate solution, a capsular membrane is formed instantaneously around each droplet due to the cross-linking of the interfacial alginate molecules with calcium cations.

Alginate capsules and particles are deposited onto a plastic petri dish in order to confirm their different morphology.

Photographs are taken of both (particles and capsules).

## 2. Synthesis of Chemically Cross-linked Gel-Building Polymer<sup>2</sup>



### Materials

|                             |                                     |
|-----------------------------|-------------------------------------|
| Acrylic acid                | 7,2 g (0,1 mol)                     |
| Butandiol diacrylate (BDAA) | 200 mg                              |
| Sodium disulfite            | 0,5 mol%, referring to acrylic acid |
| Potassium peroxodisulfate   | 0,5 mol%, referring to acrylic acid |
| Ultra-pure water            |                                     |
| Sodium chloride solution    | 100 ml (0,5 M)                      |
| Calcium chloride solution   | 100 ml (0,17 M)                     |

Tri-necked flask (100 mL)  
 Beaker flask (3 x 100 mL)  
 Reflux condenser  
 Filter paper  
 Bubble counters  
 Magnetic stirrer  
 Egg shaped magnetic stir bar  
 Glass stick  
 3x Petri dish  
 3x Watch glass  
 Glass stopper NS 14,5  
 Silicon stopper  
 1x Tweezers

<sup>2</sup> D. Braun, H. Cherdron, M. Rehahn, H. Ritter, B. Voigt. Polymer Synthesis: Theory and Practice, 5th Edition, Springer 2012, p. 378.

**Synthesis:**

Acrylic acid and 20 mL of distilled water are placed in a 100 mL flask and stirred for 15 min under nitrogen. Butandiol diacrylate (BDDA) is added to the mixture. 0,5 mol% of sodium disulfite and 0,5 mol% of potassium peroxodisulfate is dissolved in 1 mL water and added to the solution. The reaction mixture is stirred 1 h under nitrogen. The gel effect occurs after 1 h and the polymerization ends.

**Ionic strength dependent swelling:**

Synthesized hydrogels are broken into small pieces and dried under vacuum (80 °C, 2 hours). To characterize the superabsorbent polymer, first a teabag is added into distilled water. After a few minutes the teabag is removed and allowed to drain for ten minutes. The weight of the drained teabag is determined. Subsequently, three teabags with 2 g of the superabsorbent polymer are prepared. One is put into 100 mL ultrapure water the second one in 100 mL NaCl-solution (0,5 M) and the third in 100 mL of a 0,17 M CaCl<sub>2</sub>-solution. The water absorption is determined by weighing the drained teabag after one, (two, three) hours and on the next day (about 20 hours) using a balance..By the use of the followed equation the swelling degree can be defined:

$$q = \frac{(a - b)}{b}$$

q= degree of swelling, a= mass of swollen network, b= mass of dry mass

### 3. *Synthesis of poly(N-isopropylacrylamide) microgels via precipitation polymerization*

#### a) Studying the impact of initiator variation

##### Materials

|                                    |                      |
|------------------------------------|----------------------|
| N-Isopropylacrylamide (NIPAM)      | 0.7 g (6.2 mmol)     |
| N,N-Methylene(bis)acrylamide (MBA) | 0.05 g (0.325 mmol)  |
| Sodium dodecyl sulfate (SDS)       | 0.025 g (0.087 mmol) |
| Ammonium persulfate (APS)          | Different amounts*   |
| Ultrapure water                    |                      |

- \* Group A: 0.1 g (0.438 mmol)  
Group B: 0.5 g (2.191 mmol)

#### b) Studying the impact of surfactant variation

##### Materials

|                                    |                     |
|------------------------------------|---------------------|
| N-Isopropylacrylamide (NIPAM)      | 0.7 g (6.2 mmol)    |
| N,N-Methylene(bis)acrylamide (MBA) | 0.05 g (0.325 mmol) |
| Sodium dodecyl sulfate (SDS)       | Different amounts*  |
| Ammonium persulfate (APS)          | 0.2 g (0.876 mmol)  |
| Ultrapure water                    |                     |

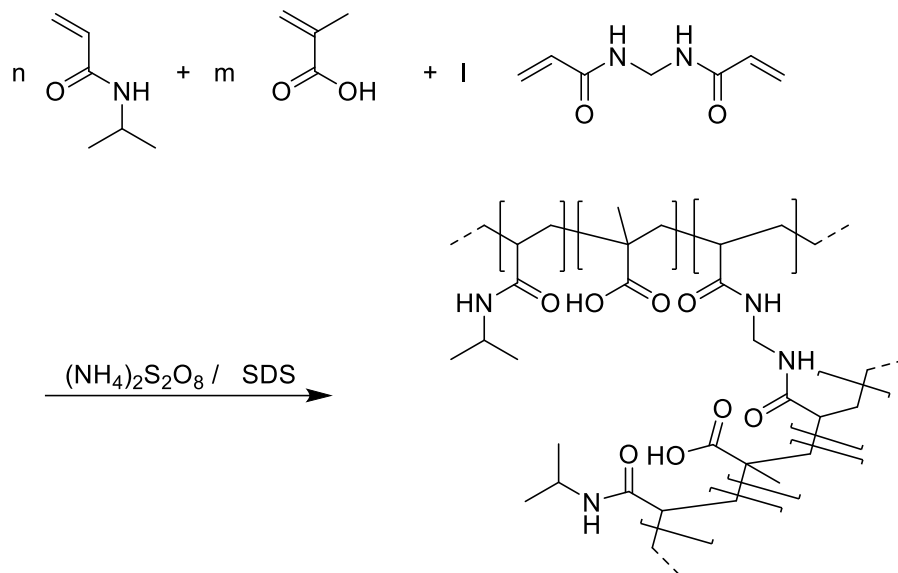
- \* Group A: 0.025 g (0.087 mmol)  
Group B: 0.050 g (0.174 mmol)

Tri-necked flask (250 mL)  
Bubble counters  
Reflux condenser  
Glass stopper  
Silicon stopper  
Magnetic stirrer  
Egg shaped magnetic stir bar  
Tapping mode cantilever

NIPAM, MBA and SDS are dissolved in 150 mL ultra-pure water. The mixture is then heated to 70°C under constant nitrogen flow for 40 minutes. The initiator (APS) is meanwhile dissolved in 10 mL ultra-pure water and injected with a syringe into the solution. The solution will start to get a blueish blur. The mixture is stirred (200 rpm) for 2 hours under constant nitrogen flow. After polymerization the solution is left to cool.

**Dynamic light scattering:** Next, 200 µL of the microgel dispersion are deposited into a plastic cuvette and filled up with 2 mL of water to perform dynamic light scattering experiment. The particle size is determined via dynamic light scattering at 20°C and 40°C as instructed by the supervisor. Note the temperature dependent swelling of the particles and their overall size in solution.

#### 4. Synthesis of poly(*N*-isopropylacrylamide)-co-methacrylic acid microgels via precipitation polymerization



##### Materials

|                                    |                      |
|------------------------------------|----------------------|
| N-Isopropylacrylamide (NIPAM)      | 0.7 g (6.2 mmol)     |
| N,N-Methylene(bis)acrylamide (MBA) | 0.05 g (0.325 mmol)  |
| Sodium dodecyl sulfate (SDS)       | 0.025 g (0.087 mmol) |
| Methacrylic acid                   | Different amounts*   |
| Ammonium persulfate (APS)          | 0.1 g (0.438 mmol)   |
| Ultrapure water                    |                      |

- \* Group A: 1 mol% (reference NIPAM + MBA)  
Group B: 10 mol% (reference NIPAM + MBA)

NIPAM, MBA and SDS are dissolved in 150 mL ultra-pure water, followed by the addition of methacrylic acid. The mixture is then heated to 70°C under constant nitrogen gas for 40 minutes. The initiator (APS) is meanwhile dissolved in 10 mL ultra-pure water and injected with a syringe into the solution. The solution will start to get a blueish blur. The mixture is stirred (200 rpm) for 2 hours under constant nitrogen flow. After polymerization the solution is left to cool and the solvent is decanted.

**Zeta Potential measurements:** Prepare the 5 mM phosphate buffer for zeta potential measurement. Centrifuge and wash at least 2 ml of the microgels with the phosphate buffer. Use a single DTS1070 folded capillary cell for the zeta potential measurements. Fill the cell with the microgels according to the supervisor's instructions and determine the zeta potential of microgels